



Formulas with Quadratics

Many formulas include squared variables. To solve for a squared variable in a formula, you can use the same techniques you use to solve quadratic equations with a single variable.

EXAMPLE 1

Solve the formula $F = \frac{GmM}{r^2}$ for r .

SOLUTION

$$F = \frac{GmM}{r^2}$$

$$r^2 F = GmM$$

Multiply each side by r^2 .

$$r^2 = \frac{GmM}{F}$$

Divide each side by F .

$$r = \pm \sqrt{\frac{GmM}{F}}$$

Take the square root of each side.

$$r = \pm \frac{\sqrt{GmM}}{\sqrt{F}} \cdot \frac{\sqrt{F}}{\sqrt{F}}$$

Rationalize the denominator.

$$r = \pm \frac{\sqrt{GmMF}}{F}$$

Simplify.

The solutions are $r = \frac{\sqrt{GmMF}}{F}$ and $r = -\frac{\sqrt{GmMF}}{F}$. ■

► **Think About It** In the formula for Example 1, F is the force of gravity between two bodies with masses m and M having centers that are separated by a distance r . G is a proportionality constant.

EXAMPLE 2

Solve $d = -16t^2 + 64t$ for t .

SOLUTION

The equation has terms of t^2 and t . You can rewrite the equation in standard form $ax^2 + bx + c = 0$, using t instead of x as the variable.

$$d = -16t^2 + 64t$$

$$0 = -16t^2 + 64t - d \quad \text{Subtract } d \text{ from each side.}$$

$$-16t^2 + 64t - d = 0 \quad \text{Standard form}$$

You can now use the quadratic formula to solve $-16t^2 + 64t - d = 0$ for t .

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{Quadratic formula}$$

$$t = \frac{-64 \pm \sqrt{64^2 - 4(-16)(-d)}}{2(-16)} \quad \text{Let } x = t, a = -16, b = 64, \text{ and } c = -d.$$

$$t = \frac{-64 \pm \sqrt{4096 - 64d}}{-32} \quad \text{Simplify.}$$

$$t = \frac{-64 \pm \sqrt{64(64 - d)}}{-32} \quad \text{Factor 64 out of both terms in the radicand.}$$

$$t = \frac{-64 \pm 8\sqrt{64 - d}}{-32} \quad \text{Simplify.}$$

$$t = \frac{8 \pm \sqrt{64 - d}}{4} \quad \text{Simplify.}$$

$$\text{The solutions are } t = \frac{8 + \sqrt{64 - d}}{4} \text{ and } t = \frac{8 - \sqrt{64 - d}}{4}. \quad \blacksquare$$

► **Think About It** In the formula for Example 2, d is the distance (in feet) above the ground for an object t seconds after it is released with an upward velocity of 64 ft/s.